

WMDR2-core

Package in package 'Model'

WMDR2-core

Version 0.2.1 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-23

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

wmdr2-core v0.2.x

v0.2.0

- time (a temporal extent) shall be realized as time:interval (unchanged from v.0.1.13)
- Following the OGC Features model, geometry = geolocation = coordinates + elevation
- All temporal<Name> elements are renamed to historical<Name>.
- historical<Name> elements other than historicalLocation shall be realized as arrays of dictionary objects for (begin) date, and the values of <Name>, e.g.,

```
historicalTerritory = [{
  "date": "2000-01-01",
  "territory": "<ISO1>"
},
{
  "date": "2024-07-13",
  "territory": "<ISO2>"
}]
```

- "observation" describes time-series (collections) of (individual) observations / measurements.
- "observation"s are acquired using a "historicalObservingSchedule", which has a (begin) date and refers to a schedule / coverage.
- "deployment" is removed as a concept, most attributes are directly under "observation". - "instrument" links directly to "observation". In alignment with OMS/STA, there is a link to "facility" via "deploymont", but this is not expected to be used at this point.
- "observation"s can be grouped to an "observationSets" (NB: This is a test and not guaranteed to exist in future.)

v0.2.1

- "observation" has been renamed to "observations", because it describes collections (=time-series) of (individual) observations / measurements.
- "observations" are acquired using a "historicalObservingSchedule", which has a (begin) date and refers to a schedule / coverage.
- The concept of deployment has been developed into "historicalDeployment", which is now a more focussed and intuitive concept. It describes the placement at a specific date of an instrument at a location (=geometry) that can generate observations. It describes the "exposure" of the instrument as well as its "operatingStatus" and "serialNumber" from that date onwards.
- "observations" can be grouped to an "observationSets" (NB: This is experimental and not guaranteed to exist in future.)
- "reporting" and "historicalReporting" are modeled to allow reuse of a reporting template. This is similar to the "schedule" / "historicalObservingSchedule" concept.

WMDR2-core

Version 0.2.1

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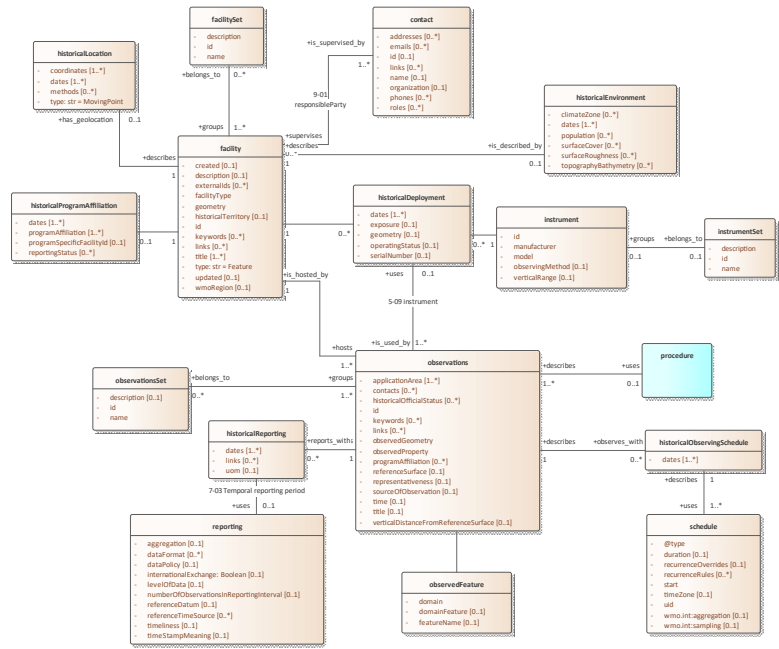


Figure 1: WMDR2-core

WMDR2-core-logical diagram

Class diagram in package 'WMDR2-core'

WMDR2core

Notes

- temporalExtent shall be realized as time:interval
- temporal<Name> elements other than temporalExtent shall be realized as an array of two arrays, one for times, one for the values of <Name>, e.g., temporalTerritory = [[2000-01-01, 2024-07-13], ["https://codes.wmo.int/wmdr/territory/<ISO1>", "https://codes.wmo.int/wmdr/territory/<ISO2>"]].
- Following the OGC Features model, geometry = geolocation = coordinates + elevation

version 0.1.4: simplify observation, specifically specification of 1-01

WMDR2-core-logical

Version 0.1.4

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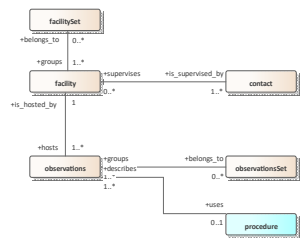


Figure 2: WMDR2-core-logical

contact

Class in package 'WMDR2-core'

WMDR2 core JSON class: contact.

contact

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
addresses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Postal addresses. [Is static True. Containment is Not Specified.]	
emails : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Email contacts. [Is static True. Containment is Not Specified.]	
id : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional stable contact identifier. [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Online resources associated with the contact. [Is static True. Containment is Not Specified.]	
name : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES	
Either a full name, or a sequence of name parts. Examples: Dr. Jörg Klausen (this corresponds to FN in vcard) Klausen;Jörg;Dr.; (this corresponds to N in vcard) [Is static True. Containment is Not Specified.]	
 organization : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Example: ORG:Federal Office of Meteorology and Climatology MeteoSwiss [Is static True. Containment is Not Specified.]	
 phones : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Phone contacts. [Is static True. Containment is Not Specified.]	
 roles : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Specific contact roles such as owner, pointOfContact, principalInvestigator. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) 9-01 responsibleParty Source: Public supervises (Class) facility Cardinality: [0..*] Target: Public is_supervised_by (Class) contact Cardinality: [1..*]	

facility

Class in package 'WMDR2-core'

Facility properties payload of the WMDR2 full record. Root Features geometry/time are modeled on fullRecord, not here.

facility

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
 created : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record creation date/time metadata. [Is static True. Containment is Not Specified.]	
 description : Private	

ATTRIBUTES	
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Facility description. [Is static True. Containment is Not Specified.]	
externalIds : Private Alias: 3-06 (primary & additional) WSIs Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) External identifiers such as WMO-WIGOS. [Is static True. Containment is Not Specified.]	
facilityType : Private Alias: 3-04 facilityType Compact WMDR facility type code value. [Is static True. Containment is Not Specified.]	
geometry : Private Alias: 3-07 current geolocation geometry = geolocation = coordinates + elevation [Is static True. Containment is Not Specified.]	
historicalTerritory : Private Alias: temporal Territory Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDS reference: 3-02 mandatory for fixed facilities, optional for mobile facilities. [Is static True. Containment is Not Specified.]	
id : Private Alias: 3-06 (primary) WSI to prefixed with name space "facility:" [Is static True. Containment is Not Specified.]	
keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Discovery/search keywords retained in the core JSON. [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links. [Is static True. Containment is Not Specified.]	
title : Private Alias: 3-03 facilityName	

ATTRIBUTES	
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) properties.title = 3-03 facilityName [Is static True. Containment is Not Specified.]	
 type : str Private = Feature Facility properties type; const facility. [Is static True. Containment is Not Specified.]	
 updated : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record update date/time metadata. [Is static True. Containment is Not Specified.]	
 wmoRegion : Private Alias: 3-01 wmoRegion Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities. [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) facility Cardinality: [1] Target: Public (Class) historicalProgramAffiliation Cardinality: [0..1]	
 Association (direction: Unspecified) Source: Public describes (Class) facility Cardinality: [1] Target: Public has_geolocation (Class) historicalLocation Cardinality: [0..1]	
 Association (direction: Unspecified) Source: Public groups (Class) facility Cardinality: [1..*] Target: Public belongs_to (Class) facilitySet Cardinality: [0..*]	
 Association (direction: Unspecified) Source: Public (Class) facility Cardinality: [1] Target: Public (Class) historicalDeployment Cardinality: [0..*]	
 Association (direction: Unspecified) Source: Public describes (Class) facility Cardinality: [1] Target: Public is_described_by (Class) historicalEnvironment Cardinality: [0..1]	

ASSOCIATIONS	
Association (direction: Unspecified) 9-01 responsibleParty	
Source: Public supervises (Class) facility Cardinality: [0..*]	Target: Public is_supervised_by (Class) contact Cardinality: [1..*]
Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observations Cardinality: [1..*]

facilitySet

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

facilitySet

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
description : Private	[Is static True. Containment is Not Specified.]
id : Private	[Is static True. Containment is Not Specified.]
name : Private Alias: title	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
Association (direction: Unspecified)	
Source: Public groups (Class) facility Cardinality: [1..*]	Target: Public belongs_to (Class) facilitySet Cardinality: [0..*]

historicalDeployment

Class in package 'WMDR2-core'

historicalDeployment

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES

ATTRIBUTES	
🔒 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
🔒 exposure : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Compact exposure classification value. [Is static True. Containment is Not Specified.]	
🔒 geometry : Private Alias: temporalGeometry Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
🔒 operatingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) values refer to http://codes.wmo.int/wmdr/InstrumentOperatingStatus [Is static True. Containment is Not Specified.]	
🔒 serialNumber : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
✓ Association (direction: Unspecified) Source: Public (Class) historicalDeployment Cardinality: [0..*] Target: Public (Class) instrument Cardinality: [1]	
✓ Association (direction: Unspecified) 5-09 instrument Source: Public is _used by (Class) observations Cardinality: [1..*] Target: Public uses (Class) historicalDeployment Cardinality: [0..1]	
✓ Association (direction: Unspecified) Source: Public (Class) facility Cardinality: [1] Target: Public (Class) historicalDeployment Cardinality: [0..*]	

historicalEnvironment

Class in package 'WMDR2-core'

Facility-environment object containing climate, surface-cover, population, and topography/bathymetry histories.

ATTRIBUTES		
 climateZone : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	climate-zone history.	[Is static True. Containment is Not Specified.]
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)		[Is static True. Containment is Not Specified.]
 population : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	History of population in 10 and 50 km radius around the facility	[Is static True. Containment is Not Specified.]
 surfaceCover : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	surface-cover history.	[Is static True. Containment is Not Specified.]
 surfaceRoughness : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)		[Is static True. Containment is Not Specified.]
 topographyBathymetry : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)		[Is static True. Containment is Not Specified.]
ASSOCIATIONS		
 Association (direction: Unspecified)		
Source: Public describes (Class) facility Cardinality: [1]	Target: Public is_described_by (Class) historicalEnvironment Cardinality: [0..1]	

historicalLocation

Class in package 'WMDR2-core'

ATTRIBUTES	
coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation]. [Is static True. Containment is Not Specified.]	
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates. [Is static True. Containment is Not Specified.]	
methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod [Is static True. Containment is Not Specified.]	
type : str Private = MovingPoint [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public describes (Class) facility Cardinality: [1] Target: Public has_geolocation (Class) historicalLocation Cardinality: [0..1]	

historicalObservingSchedule

Class in package 'WMDR2-core'

When are observations made? Describes a temporal schedule/coverage, e.g. "from Feb-Oct on working days from 06-18 UTC".
Objects like

{"date": date, "schedule": str}, whereby "schedule" is a reference to a re-usable schedule object.

historicalObservingSchedule

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with observingSchedule. [Is static True. Containment is Not Specified.]

ASSOCIATIONS	
✍ Association (direction: Unspecified)	
Source: Public describes (Class) historicalObservingSchedule Cardinality: [1]	Target: Public uses (Class) schedule Cardinality: [1..*]
✍ Association (direction: Unspecified)	
Source: Public describes (Class) observations Cardinality: [1]	Target: Public observes_with (Class) historicalObservingSchedule Cardinality: [0..*]

historicalProgramAffiliation

Class in package 'WMDR2-core'

historicalProgramAffiliation

Version 1.0 Phase 1.0 Proposed

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Alias temporalProgramAffiliation

ATTRIBUTES	
🔒 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	
Date array aligned with programAffiliation.	[Is static True. Containment is Not Specified.]
🔒 programAffiliation : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	
Compact program-affiliation values.	[Is static True. Containment is Not Specified.]
🔒 programSpecificFacilityId : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
	[Is static True. Containment is Not Specified.]
🔒 reportingStatus : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
http://codes.wmo.int/wmdr/ReportingStatus	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
✍ Association (direction: Unspecified)	
Source: Public (Class) facility Cardinality: [1]	Target: Public (Class) historicalProgramAffiliation Cardinality: [0..1]

ASSOCIATIONS

historicalReporting

Class in package 'WMDR2-core'

historicalReporting

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ATTRIBUTES
🔗 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]
🔗 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) URL to related WIS record, entry point for data discovery and retrieval. [Is static True. Containment is Not Specified.]
🔗 uom : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/TimeStampMeaning. [Is static True. Containment is Not Specified.]

ASSOCIATIONS
🔗 Association (direction: Unspecified) 7-03 Temporal reporting period Source: Public describes (Class) historicalReporting Cardinality: [1] Target: Public uses (Class) reporting Cardinality: [0..1]
🔗 Association (direction: Unspecified) Source: Public describes (Class) observations Cardinality: [1] Target: Public reports_ with (Class) historicalReporting Cardinality: [0..*]

instrument

Class in package 'WMDR2-core'

Describes instrument capabilities, not a specific instrument instance.
For semantic consistency, terms like observedProperty and observedGeometry are used, even though they just describe the instrument capabilities, not the actual observations. With this, instrument becomes re-usable between deployments and even facilities.

ATTRIBUTES	
 id : Private [Is static True. Containment is Not Specified.]	
 manufacturer : Private Alias: 5-09 manufacturer [Is static True. Containment is Not Specified.]	
 model : Private Alias: 5-09 model [Is static True. Containment is Not Specified.]	
 observingMethod : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) from (new) unified code list of observing methods, based on http://codes.wmo.int/wmdr/ObservingMethodAtmosphere http://codes.wmo.int/wmdr/ObservingMethodTerrestrial http://codes.wmo.int/wmdr/ObservingMethodEarth [Is static True. Containment is Not Specified.]	
 verticalRange : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public groups (Class) instrument Cardinality: [0..1] Target: Public belongs_to (Class) instrumentSet Cardinality: [0..1]	
 Association (direction: Unspecified) Source: Public (Class) historicalDeployment Cardinality: [0..*] Target: Public (Class) instrument Cardinality: [1]	

instrumentSet

Class in package 'WMDR2-core'

ATTRIBUTES	
 description : Private	[Is static True. Containment is Not Specified.]
 id : Private	[Is static True. Containment is Not Specified.]
 name : Private Alias: title	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public groups (Class) instrument Cardinality: [0..1]	Target: Public belongs_to (Class) instrumentSet Cardinality: [0..1]

observations

Class in package 'WMDR2-core'

WMDR2 core JSON class: observation.

observations

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ATTRIBUTES	
 applicationArea : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Value(s) from http://codes.wmo.int/wmdr/ApplicationArea	[Is static True. Containment is Not Specified.]
 contacts : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation-level contacts.	[Is static True. Containment is Not Specified.]
 historicalOfficialStatus : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) [{"officialStatus": str, "date": date}, {...}], whereby "officialStatus" values refer to http://codes.wmo.int/wmdr/ObservationStatus	[Is static True. Containment is Not Specified.]
 id : Private	

ATTRIBUTES	
Observation identifier, e.g. observation:<observedVariable>.	[Is static True. Containment is Not Specified.]
keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation keywords.	[Is static True. Containment is Not Specified.]
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation links.	[Is static True. Containment is Not Specified.]
observedGeometry : Private cf. http://codes.wmo.int/wmdr/Geometry	[Is static True. Containment is Not Specified.]
observedProperty : Private Alias: OMS:ObservedProperty, iop:Variable The <code>observedVariable</code> is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2. NB: <code>observedVariable</code> will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the <code>observedProperty</code>, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific. Examples include: - <code>observableProperty</code>: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...) - <code>objectOfInterest</code>: sea ice, CO2, aerosol particles, chlorophyll - <code>observablePropertyConstraints</code>: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text. - <code>matrix</code>: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ... These elements will be supported by code lists from codes.wmo.int/wmdr2. Applications may want to guide users to select a specific <code>observedVariable</code> by offering groupings, filtering, or tree-like visualizations according to matrix, <code>objectOfInterest</code>, etc.	[Is static True. Containment is Not Specified.]
programAffiliation : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Program-affiliation timeline fragments inherited from WMDR1 observations.	[Is static True. Containment is Not Specified.]
referenceSurface : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES	
Linked to heightAboveReferenceSurface, from http://codes.wmo.int/wmdr/ReferenceSurfaceType [Is static True. Containment is Not Specified.]	
representativeness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/Representativeness [Is static True. Containment is Not Specified.]	
sourceOfObservation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/SourceOfObservation. If sourceOfObservation=="humanObservation", several attributes of "observations" are expected to be null. [Is static True. Containment is Not Specified.]	
time : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Observation temporal extent, date-resolution interval or null. [Is static True. Containment is Not Specified.]	
title : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Human-readable title derived from domain, geometry, observed variable, and optional label. [Is static True. Containment is Not Specified.]	
verticalDistanceFromReferenceSurface : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) implement as an array of objects: [{ "value": float, "uom": str }, { ..., .. }], whereby "uom" from http://codes.wmo.int/wmdr/unit [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public groups (Class) observations Cardinality: [1..*] Target: Public belongs_to (Class) observationsSet Cardinality: [0..*]	
Association (direction: Unspecified) Source: Public (Class) observations Target: Public (Class) observedFeature	
Association (direction: Unspecified) 5-09 instrument Source: Public is_used_by (Class) observations Cardinality: [1..*] Target: Public uses (Class) historicalDeployment Cardinality: [0..1]	

ASSOCIATIONS	
Association (direction: Unspecified) 3-07 temporalGeometry	
Source: Public (Class) observations Cardinality: [1]	Target: Public (Class) temporalGeometry Cardinality: [0..1]
describes	has_geolocatino
Association (direction: Unspecified)	
Source: Public describes (Class) observations Cardinality: [1]	Target: Public reports_with (Class) historicalReporting Cardinality: [0..*]
Association (direction: Unspecified)	
Source: Public describes (Class) observations Cardinality: [1..*]	Target: Public uses (Class) procedure Cardinality: [0..1]
ObservingProcedure(s)	
Association (direction: Unspecified)	
Source: Public describes (Class) observations Cardinality: [1]	Target: Public observes_with (Class) historicalObservingSchedule Cardinality: [0..*]
Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observations Cardinality: [1..*]

observationsSet

Class in package 'WMDR2-core'

grouping of observations (which are themselves time series)

observationsSet

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES
description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Deployment temporal extent, date-resolution interval or null. [Is static True. Containment is Not Specified.]
id : Private

ATTRIBUTES	
Deployment identifier; source XML id is preserved when available.	[Is static True. Containment is Not Specified.]
 name : Private	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public groups (Class) observations Cardinality: [1..*]	Target: Public belongs_to (Class) observationsSet Cardinality: [0..*]

observedFeature

Class in package 'WMDR2-core'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-feature-of-interest, of an observation.

observedFeature

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
 domain : Private	
The largest scale in WIGOS, cf. http://codes.wmo.int/wmdr/Domain	[Is static True. Containment is Not Specified.]
 domainFeature : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Describes the type of the larger-scale environmental target or ultimate-feature-of-interest, of an observation.	
Concept motivated by requirements from hydrology, but may be useful elsewhere. New code list is proposed, with terms like	
- river basin	
- lake	
- river	
- catchment	
- canopy	
- epilimnion	
- hypolimnion	
- troposphere	
- tropopause	
- stratosphere	
- thermocline (useful as a general term, but needs "domain")	
- boundary layer (useful as a general term, but needs "domain")	

ATTRIBUTES	
- ... [Is static True. Containment is Not Specified.]	
featureName : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) free text field, e.g., "Danube Basin", "Amazon rain forest", "Amazon river Basin" [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) observations Target: Public (Class) observedFeature	

procedure

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

procedure

Version 1.0 Phase 1.0 Proposed

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ASSOCIATIONS	
Association (direction: Unspecified) Source: Public describes (Class) observations Cardinality: [1..*] Target: Public uses (Class) procedure Cardinality: [0..1] ObservingProcedure(s)	

reporting

Class in package 'WMDR2-core'

Reusable reporting template.

reporting

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
aggregation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Reporting/aggregation period, renamed from temporalReportingInterval. [Is static True. Containment is Not Specified.]	

ATTRIBUTES	
dataFormat : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/DataFormat. [Is static True. Containment is Not Specified.]	
dataPolicy : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/DataPolicy. [Is static True. Containment is Not Specified.]	
internationalExchange : Boolean Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
levelOfData : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/LevelOfData. [Is static True. Containment is Not Specified.]	
numberOfObservationsInReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
referenceDatum : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/CoordinateReferenceSystem. [Is static True. Containment is Not Specified.]	
referenceTimeSource : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Single value or array, values from http://codes.wmo.int/wmdr/ReferenceTime. [Is static True. Containment is Not Specified.]	
timeliness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Expected timeliness of reporting, an object with value and uom (default: seconds) [Is static True. Containment is Not Specified.]	
timeStampMeaning : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/TimeStampMeaning. [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
 Association (direction: Unspecified) 7-03 Temporal reporting period	
Source: Public describes (Class) historicalReporting Cardinality: [1]	Target: Public uses (Class) reporting Cardinality: [0..1]

schedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

schedule

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ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
 @type : Private	
JSCalendar object type; const Event.	[Is static True. Containment is Not Specified.]
 duration : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Occurrence duration describing coverage window or sampling duration.	[Is static True. Containment is Not Specified.]
 recurrenceOverrides : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
JSCalendar recurrence override patches; null override values are invalid.	[Is static True. Containment is Not Specified.]
 recurrenceRules : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
JSCalendar recurrence rules.	[Is static True. Containment is Not Specified.]
 start : Private	
Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start.	[Is static True. Containment is Not Specified.]
 timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
IANA time zone or UTC default.	

ATTRIBUTES	
	[Is static True. Containment is Not Specified.]
uid : Private JSCalendar-safe schedule uid.	[Is static True. Containment is Not Specified.]
wmo.int:aggregation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDR-specific aggregation metadata.	[Is static True. Containment is Not Specified.]
wmo.int:sampling : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDR-specific sampling metadata preserved from dataGeneration.sampling.	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public describes (Class) historicalObservingSchedule Cardinality: [1] Target: Public uses (Class) schedule Cardinality: [1..*]	

temporalGeometry

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalGeometry.

temporalGeometry

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-12. Last modified 2026-06-02

ATTRIBUTES	
coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation].	[Is static True. Containment is Not Specified.]
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates.	[Is static True. Containment is Not Specified.]

ATTRIBUTES	
methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
use http://codes.wmo.int/wmdr/GeopositioningMethod	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
Association (direction: Unspecified) 3-07 temporalGeometry	
Source: Public (Class) observations Cardinality: [1]	Target: Public (Class) temporalGeometry Cardinality: [0..1]
describes	has_geolocatino